



Semantic process benchmarking to improve process performance

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Abstract

Purpose – The purpose of this paper is to illustrate and evaluate the semantic process benchmarking concept.

Design/methodology/approach – The authors' approach includes the use of metamodels and ontologies, which make the process models syntactically and semantically comparable. Furthermore, a software prototype is presented to analyze and compare individual process models and their performance information. Thereafter, the technical, conceptual, and economic perspectives of the approach's evaluation are aligned with their respective outcomes.

Findings – The evaluation proves that this approach is generally suitable to generate novel and useful information on different process models and their performance within the same problem domain. However, the initial set-up costs are high and will only pay off once process models are used regularly.

Practical implications – The proposed approach depends strongly on the availability of appropriate metrics and ontologies, as well as on the annotation of these ontologies to process models, which is a time-consuming task. If large benchmarking clearing centers are established, the approach will be more cost-effective. The developed SEMAT prototype, that demonstrates and proves the proposed approach's general viability, supports cost-effective ontology engineering and annotation in the context of semantic process benchmarking initiatives.

Originality/value – To date, process benchmarking has primarily been a manual process. In this article, the authors suggest an approach that allows time-consuming and costly process analysis to be partially automated, which makes the performance indicators, as well as qualitative differences between processes, apparent.

Keywords Benchmarking, Process analysis, Process modeling, Business process management, Ontologies, Metamodeling, Semantic model analysis, Semantic benchmarking

Paper type Research paper



1. Introduction

Organizations use benchmarking widely to identify areas in which they can improve their efficiency (Watson, 1993). An important prerequisite for improved efficiency is capturing the data of key performance indicators (KPIs) (Alshawi *et al.*, 2003). In their quest for such data, benchmarkers have mainly focused on products and services, and have rarely included processes (Ahmed and Rafiq, 1998; Carpinetti and Melo, 2002).

In general, two different types of benchmarking can be distinguished: “quantitative approaches” are based on performance indicators, while “qualitative approaches” make use of in-depth analyses of the organizations to be benchmarked. The latter approaches involve interviewing, document analysis, and observation. Qualitative data gathering can uncover performance gaps, and can be used to investigate the causes of higher or lower performance. Best practices can also be identified and made available to the benchmarking participants. However, qualitative approaches require a significant number of analyses, and depend on experienced and skilled benchmarking experts. As Drew pointed out in 1997, new ways of making benchmarking less time-consuming and costly need to be identified (Drew, 1997, p. 439). Although this call for new approaches was made several years ago, no answer has yet been provided.

To date, process benchmarking has primarily been a manual process, requiring individuals to collect data and analyze them. Experts are needed to collect and interpret the benchmarking information as it is mostly unstructured. However, even if semi-formal or formal process models are used, diverging terminologies and domain knowledge prohibit a direct analysis. Moreover, processes are often described in different process models that present dissimilar process views at diverse levels of abstraction (Markovic *et al.*, 2009). A model’s objectives and focus, as well as the choice of modeling language (Recker *et al.*, 2009) depend on the particular view of the process that the model illustrates. With extra-organizational benchmarking, a comparison can only be made if several process models, each using a different language, are employed. A careful translation into a common language, in order to gain a universal understanding, is crucial for a meaningful comparison. Owing to this complication, software-based approaches to benchmarking mainly focus on performance indicators; they do not generally involve a qualitative analysis of the data gathered.

Information systems (IS) research provides a solution to solve the problem of low benchmarking efficiency by suggesting the use of conceptual process models. As long as the process models are stored in a digital semi-formal or formal format, a partially automated analysis and comparisons can be made by using IS. However, there are two fundamental problems with this solution: First, conceptual process models are often syntactically incompatible, since they make use of different process modeling languages (e.g. event-driven process chains (EPCs), Petri nets, unified modeling language (UML) activity diagrams). Second, the terminology or semantics of common models differs (Teuteberg *et al.*, 2009). Therefore, a comparison is not possible (Thomas and Fellmann, 2009).

The objective of our research is to propose an approach that partly allows automating the time-consuming and costly analysis and comparison of process models, which will not only allow diverging performance indicators to become obvious, but also the qualitative differences between processes. This approach is based on using metamodels and ontologies for the analysis. We furthermore evaluate the proposed approach in terms of its technical, conceptual and economic perspectives. We have developed a software system prototype to specifically address the first two evaluation perspectives.

To achieve our research goal, we proceed as follows: Section 2 outlines the theoretical foundations of our work, and summarizes the approaches and results of related work. The research methodology will be described in Section 3. In Section 4, the developed method for the analysis and comparison of individual process models is presented. We deal with the threefold evaluation of our research results in Section 5. We then provide an overview of our evaluation methodology before describing the

technical, conceptual, and economic evaluation perspectives with their respective outcomes. In Section 6, we summarize the results of our work, discuss its implications and limitations (e.g. regarding ease of use and applicability), and point to the need for further research.

2. Semantic process benchmarking: the state of research

2.1 Foundations

Over the decades, benchmarking has developed in several stages. Whereas early attempts were mainly product-oriented, contemporary approaches are global and focus on processes and strategies (Ahmed and Rafiq, 1998; Kyrö, 2003). Today, benchmarking can be considered a mature field of research (Müller *et al.*, 2010). According to Dattakumar and Jagadeesh (2003, p. 185), more than 350 publications related to benchmarking were available in June 2002. The generic benchmarking concept is applied in almost all contemporary enterprises' functional areas (Dattakumar and Jagadeesh, 2003, p. 190).

While there is still no consistent theoretical definition of benchmarking (Moriarty and Smallman, 2009), there are many classifications of benchmarking approaches. According to Carpinetti and Melo (2002, p. 245), benchmarking can be classified according to its object. Consequently, they distinguish between product, process, and strategic benchmarking. A wide spectrum of methods is available for benchmarking, especially for process benchmarking (Francis and Holloway, 2007; Legner, 1999, pp. 31-82). Although these approaches vary in detail, they have several fundamental characteristics in common. Five key benchmarking process activities can be identified (Drew, 1997, p. 430):

- (1) determine what to benchmark;
- (2) form a benchmarking team;
- (3) identify benchmarking partners;
- (4) collect and analyze benchmarking data; and
- (5) take action.

If this process is followed, benchmarking is an excellent tool for capturing rich datasets and relevant KPIs. Müller *et al.* (2010) show that the prerequisites for effective and efficient process benchmarking are instruments, databases, methodological excellence, and benchmarking data quality. According to them, the analysis of KPIs, as well as of contextual, and competitive information is an important foundation for process benchmarking success. However, subject matter experts mostly undertake benchmarking initiatives' data collection and analysis manually.

The business process management (BPM) research community has yielded a wealth of languages and standardization approaches that aim to describe and represent business processes for benchmarking initiatives. These include the EPC, the UML activity diagram, the Petri net, and the business process modeling notation (BPMN) (Thomas and Fellmann, 2009, p. 438). In the evaluation of our approach, we refer to the widely used modeling languages EPCs and UML activity diagrams. EPCs and activity diagrams visually describe the modeled process's dynamic aspects, the elemental actions' interconnection, and their integration into control and data flows.

Metamodels (Strahringer, 1998, p. 23) and ontologies play an important role in our approach to semantic process benchmarking, because they help overcome the two

fundamental problems outlined above: They make the non-compatible process model notations (e.g. EPCs, UML, BPMN, Petri nets), the modeling language constructs, and the modeled concepts both syntactically and semantically comparable.

Metamodels enable us to analyze and verify modeling languages' syntactical completeness and expressiveness by making sense of the relationships between the concepts used in different modeling languages (Karagiannis and Höfferer, 2006). These metamodels describe the modeling constructs and show how they can be connected to one another (i.e. syntax). One of metamodels' primary functions is to describe the language constructs and element types of the individual models represented in different modeling notations. Metamodels themselves can be subject to meta-modeling. Consequently, they can be arranged in hierarchies, and higher level metamodels (i.e. meta²-models) can describe the modeling constructs of lower level metamodels (Strahinger, 1998).

Since the beginning of the 1990s, research on ontologies has increased. The importance of this research has been recognized in a multiplicity of research fields and IS discipline application areas (Guarino, 1998). The Semantic Web field is one that is highly influenced; ontologies are used to structure its underlying knowledge and enable comprehensive and transportable machine understanding (Maedche and Staab, 2001). More generally, ontologies are used in the IS discipline to formally represent knowledge. They facilitate knowledge sharing and reuse between various agents, regardless of whether they are human or artificial (Fensel, 2004, p. 4).

A frequently quoted definition of an ontology within the IS discipline is (Gruber, 1993): "An ontology is a formal explicit specification of a conceptualization". Over time, this definition has been extended to: "an ontology is a [...] shared conceptualization for a domain of interest". This definition contains the three constitutive ontology characteristics relevant for the approach proposed in this article: an ontology should be machine readable ("formal"); the types of ontology concepts and their constraints should be defined "explicitly"; and an ontology should capture consensual knowledge accepted in a group ("shared").

2.2 Related work

In recent years, the synthesis of the two research areas mentioned in Section 2.1 – process benchmarking on the one hand and metamodels/ontologies on the other – has gained much attention.

To build this paper on a solid base, we applied the method of a concept-centric systematic literature review (Webster and Watson, 2002). As a first step, we defined the review scope and concentrated on the application of ontologies and metamodels in process benchmarking and process analyses. Key words for the search belong to the realm of semantic process management and include the terms "semantic", "ontology", "meta*", and "model*" combined with "benchmarking" and/or "process". The applied wildcards assure the identification of related, conjugated terms. Next, we applied these key words to well-known (scientific) search engines like SCIRUS, Google, and Google Scholar, and scientific databases like EBSCO, Science Direct, SpringerLink, and AISel to retrieve scientific, peer-reviewed papers. To enlarge the number of papers, we used forward (review of reference lists) and backward search (author-centric review).

The potential to annotate metamodels and ontologies to compare process models has been recognized and discussed in the literature for quite some time now (Hepp and Roman, 2007; Hepp *et al.*, 2005; Lin and Strasunskas, 2005; Thomas and Fellmann, 2009). The research works on semantic process modeling (Heinrich *et al.*, 2008;

Thomas and Fellmann, 2007) – a synthesis of process modeling and formal ontologies – also discuss certain aspects of individual process models’ comparison to some extent. Certain studies in the literature can be compared to our study, although they differ in specific ways. These studies are summarized in Table I, and are compared to our approach by means of the following criteria, which clarify the differences:

- *Semantic approach.* Does the study employ a semantic approach to analyze the processes?
- *Benchmarking approach.* Does the study propose a process benchmarking approach to analyze and compare individual process models efficiently and effectively?
- *Model comparison.* Does the study address the issue of comparing process models in different notations (e.g. EPC, UML, BPMN, Petri nets) containing context information by means of metamodels and ontologies?
- *Evaluation approach.* Do the authors present an evaluation approach and, if so, what kind of evaluation?
- *Problem domain and major difference.* On which problem domain does the study focus? What is the major difference between this study and our approach?

According to the criteria in Table I, the approach taken by Liao *et al.* (2005) seems to align with our approach. The authors propose an ontology-based method to describe software processes conceptually. One of their approach objectives is to quantitatively analyze and benchmark software process attributes. However, the major differences between their approach and ours lies in their specific focus on software process models, as well as that they do not map different process models; that is, they address only one specific process model at a time. Similarly, the approaches taken by Ehrig *et al.* (2007), Chanda *et al.* (2010), Delfmann *et al.* (2011) and Höfferer (2007) fail to provide concepts or methods to benchmark process models. The other studies summarized in Table I differ from our approach in at least two significant ways. In addition, they mostly focus on a different, or even limited, problem domain, such as EPCs, Petri nets or BPMNs (Daneva *et al.*, 1996; Thomas and Fellmann, 2007; Dijkman *et al.*, 2008), reference models (Fettke and Loos, 2007), software process models (Liao *et al.*, 2005), and/or pursue a different objective, for example, planning process models (Heinrich *et al.*, 2008).

3. Cyclic research process

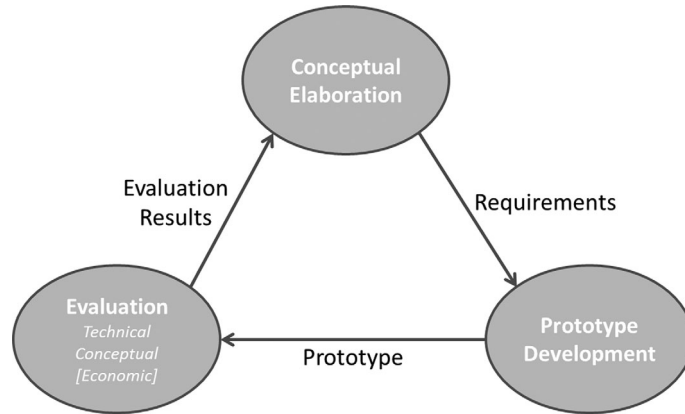
The research described in this article can be characterized as design-oriented research that focuses on the creation of a “new” method (i.e. “semantic process benchmarking using metamodels and ontologies”). Valid design science research requires an iterative search for an adequate solution to a relevant problem, and a rigorous evaluation of the solution (Walls *et al.*, 1992; March and Smith, 1995; Hevner *et al.*, 2004). Following the framework proposed by March and Smith, we follow a process in which we design and evaluate our proposed method (Figure 1). For this purpose, we create an instantiation of a software prototype (Section 5.1). However, the prototype does not constitute a major research result, but serves as a tool for evaluating our method. We expect this instantiation to:

- convey our method’s ease of use and applicability; and to
- prove the general viability of the method, which cannot be executed efficiently without appropriate software support (March and Smith, 1995, p. 260).

References	Semantic approach	Benchmarking approach	Model comparison	Evaluation approach	Problem domain and major difference to our approach
Chabrol <i>et al.</i> (2006)	No	No	No	N/A	Focuses on a methodological approach to supply chain modeling and evaluates processes in the health care system
Chanda <i>et al.</i> (2010)	Yes	No	Yes	N/A	Focuses on the verification of UML diagrams
Daneva <i>et al.</i> (1996)	No	Yes	Yes	Prototype under development	Focuses on controlling the quality of business process models
Delfmann <i>et al.</i> (2011)	Yes	No	Yes	N/A	On the basis of EPCs and an extension of the ARIS toolset
Dijkman <i>et al.</i> (2008)	Yes	No	No	Prototype	Focuses on semantic standardization and structural pattern matching in conceptual models
Ehrig <i>et al.</i> (2007)	Yes	No	Yes	Prototype	Focuses on mapping the business process modelling notation to Petri nets
El Kharbili and Stein (2008)	Yes	No	No	N/A	Focuses on measuring the similarities between semantic business process models by (semi-) automatically detecting synonyms and homonyms in process element names
Fettke and Loos (2007) and Fettke and Loos (2003)	Yes	No	Yes (proposed)	N/A	Focuses on policy-based semantic compliance checking
Filipowska <i>et al.</i> (2009)	Yes	No	No	N/A	Focuses on evaluating reference models
Heinrich <i>et al.</i> (2008)	Yes	No	No	Prototype	Focuses on ontologically representing the business perspective
Hepp and Roman (2007)	Yes	No	No	N/A (planned)	Focuses on partially automatically planning and modeling process models
Höfner (2007)	Yes	No	Yes	Prototype under development	Focuses on ontologically representing the business perspective, as well as the information systems perspective on business processes, and on translating between these two perspectives
Huth <i>et al.</i> (2001)	Yes	No	No	Prototype	Focuses on the semantic interoperability of business processes
Liao <i>et al.</i> (2005)	Yes	Yes	No	Prototype	Focuses on assessing process instances to derive process models in groupware environments
Lin and Strassnig (2005)	Yes	No	No	N/A	Focuses on software processes and their models
Thomas and Fellmann (2007)	Yes	No	No	N/A	Focuses on the storage and retrieval of process models for reuse
Thomas and Fellmann (2009), Fellmann <i>et al.</i> (2011)	Yes	No	No	N/A	Focuses on a semantic annotation of EPCs
					Focuses on a semantic annotation of EPCs to specify the semantics of individual process model elements

Table I.
Summary of related work

Figure 1.
Cyclic research process



We start our research with an in-depth analysis of related work (Section 2.2) and a deductive elaboration of the method (Section 4) outlined in this article. Subsequent to this construction phase, we translate the method into a software system's requirements and develop the prototype accordingly. The prototype is then used to perform the evaluation (Section 5). We apply three evaluation perspectives: The first perspective is conceptual, and investigates whether the proposed method works and yields useful results. The second perspective is technical, and considers the method's technological feasibility. The third perspective investigates its economic potential. In order to evaluate economic potential, we perform a descriptive and argumentative evaluation (Frank, 2006, pp. 33-35). This evaluation illustrates that certain institutional preconditions have to be met for our approach to be profitable in a real-world scenario. Sections 5 and 6 provide more information on the evaluation perspectives. At the end of each cycle, we use the evaluation results to further improve our approach. Hevner *et al.* (2004) differentiate between different evaluation strategies, i.e. observation, analysis, experimentation, testing, and describing: our approach is evaluated by developing a software prototype that we apply in simulation experiments with existing process models. We use prominent reference process models for procurement processes (Scheer, 1994; Schuh, 2006). This analysis allows us to see whether our approach is generally suitable to generate novel and useful information about different process models that describe the same problem domain.

4. Ontology-based process model benchmarking

4.1 Ontology-based benchmarking process

As pointed out in Section 1, a comparison of process models and their underlying process information for benchmark purposes leads to two fundamental problems (Teuteberg *et al.*, 2009):

- First, different modeling languages have various process views, and these may be based on non-compatible notations (syntactical aspect).
- Second, the semantics in the modeling language constructs, for example, process steps and control flows (semantical aspect I) and in the modeled concepts, for example "producing" or "manufacturing" (semantical aspect II) may differ. The latter may result from different methods of abstraction, detailing, typing, and naming (Pfeiffer and Gehlert, 2005, p. 115).

In order to overcome such problems, ontologies can be used to capture all relevant process model aspects. Furthermore, they can be used to analyze and evaluate these aspects. Ontologies have the potential to translate language constructs from one conceptual process modeling language into another language (syntactical aspect), and to explicate language construct semantics. Moreover, ontologies make it possible to identify equivalent language constructs (semantical aspect I), to represent general domain knowledge, and to semantically map elements from one process model to those of another (semantical aspect II) (Thomas and Fellmann, 2009).

For the ontology-based process benchmarking and the comparison of process models for benchmarking purposes, we will consider the following general process, and describe each step in detail.

Step 1: specifying a domain ontology. A domain expert either creates or selects a domain ontology that describes the benchmarking domain. Each stakeholder (e.g. modeling experts or a benchmarking expert) has to agree on this domain ontology due to its central relevance for the next benchmarking steps.

Step 2: create or use a base ontology. A modeling expert creates a base ontology. This base ontology must contain concepts that represent common process modeling language constructs used to model the underlying process views and their semantics. If necessary, this can be done with a reference metamodel that has already been used to translate process models from one language into another. In most cases, existing base ontologies can be reused as they have the required fundamental characteristics. This base ontology overcomes the problems related to the syntactical aspect, and semantical aspect I.

For instance, the base ontology presented in Figure 2 is a translation of a reference metamodel into an ontology. This reference metamodel indicates a potential method for standardizing and describing language process modeling language constructs (syntactical aspect and semantical aspect I). Consequently, concepts that represent the standardized and supposedly relevant element types for activity-oriented benchmarking become part of the base ontology. For example, a modeled process event is described as a subconcept of the concept “Process Element”. Through the use of the base ontology, a syntactical analysis can be undertaken of the transferred model constructs, albeit on a higher level of abstraction.

Step 3: mapping process elements to the domain ontology. A domain expert maps the elements of the process models to be benchmarked to instances of the domain ontology. This overcomes semantical aspect II, because each process step’s semantics is specified using a unified vocabulary.

In contrast to Höfferer (2007), who only describes exact mappings between ontology concepts and model constructs, this concept allows three mapping types to be used:

- (1) The model construct represents the domain concept (“isEquivalentConceptOf”) exactly.
- (2) The model construct represents the mapped domain concept in a broader sense (“isBroaderConceptOf”).
- (3) The model construct represents the mapped domain concept in a narrower sense (“isNarrowerConceptOf”).

Step 4: capture performance information. In respect of benchmarking initiatives, process model elements require capturing additional performance information, context information and/or process aspects, which are then inserted into the ontology.

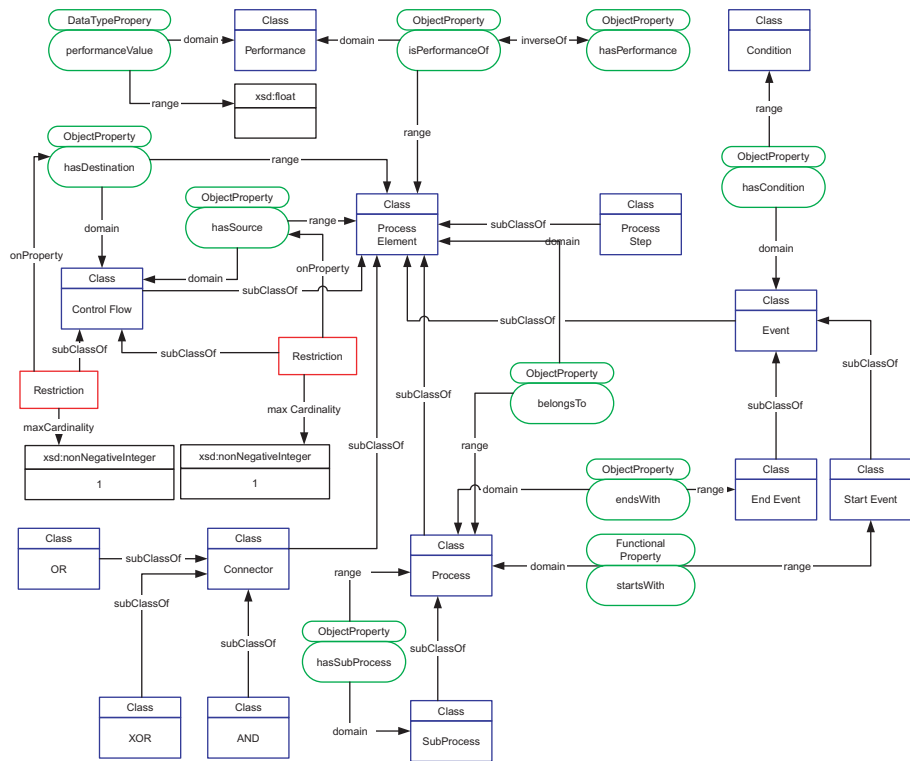


Figure 2.
Base ontology

Step 5: transform all information into a single instance ontology. A software system transforms the process models into an instance ontology of the base ontology. The system uses the mappings done in step 3 to map these instances to the domain ontology instances. Figure 3 shows the relation between process models and metamodels, as well as their transformation on the one hand, and ontologies on the other.

The application domain to be analyzed can be syntactically represented by $1, \dots, n$ models in different modeling languages (e.g. in UML or in EPC notation). Semantically, the application domain can be described by a domain ontology (semantical aspect I). According to Strahringer (1998, pp. 3-4), the models that represent an application domain by means of an object or modeling language are situated on the lowest description level (layer 0). Likewise, domain ontologies may also be allocated to the lowest description level, since they describe a restricted segment of reality by means of an ontology language. On the next higher level (layer 1), the individual object languages' metamodels – a common reference metamodel of these metamodels – and the base ontology, which is an equivalent representation of a reference metamodel, are presented. The “base ontology” represents the terminological knowledge of the “analysis ontology”, i.e. it provides the reference metamodel's elementary concepts. Figure 4 shows the XML representation (based on Event-Driven-Process-Chain-Markup-Language (EPML) and Web Ontology Language (OWL)) of a model construct (“Inventory Management”).

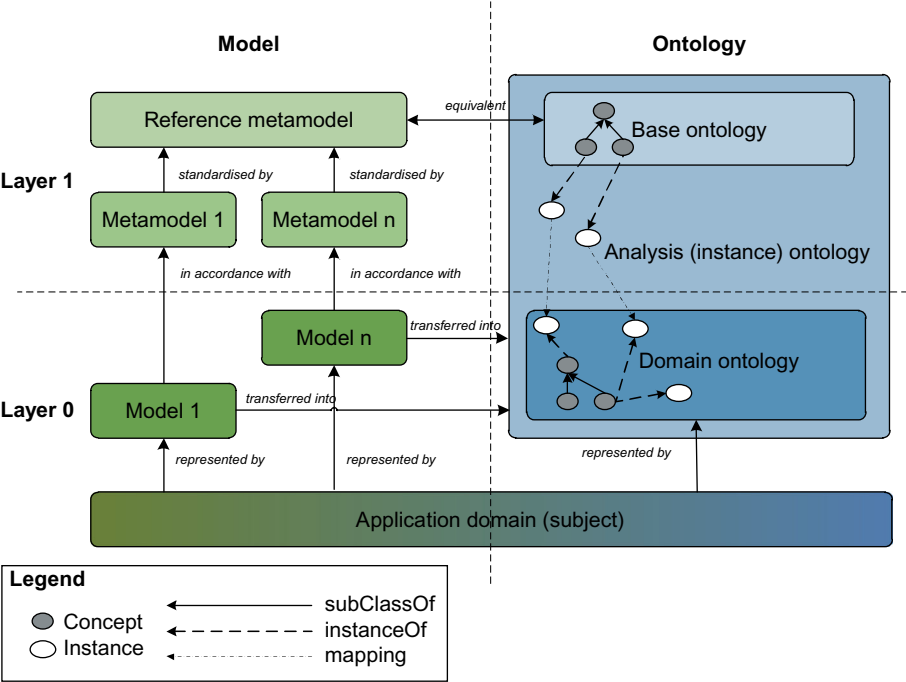


Figure 3. Syntactic and semantic coherence

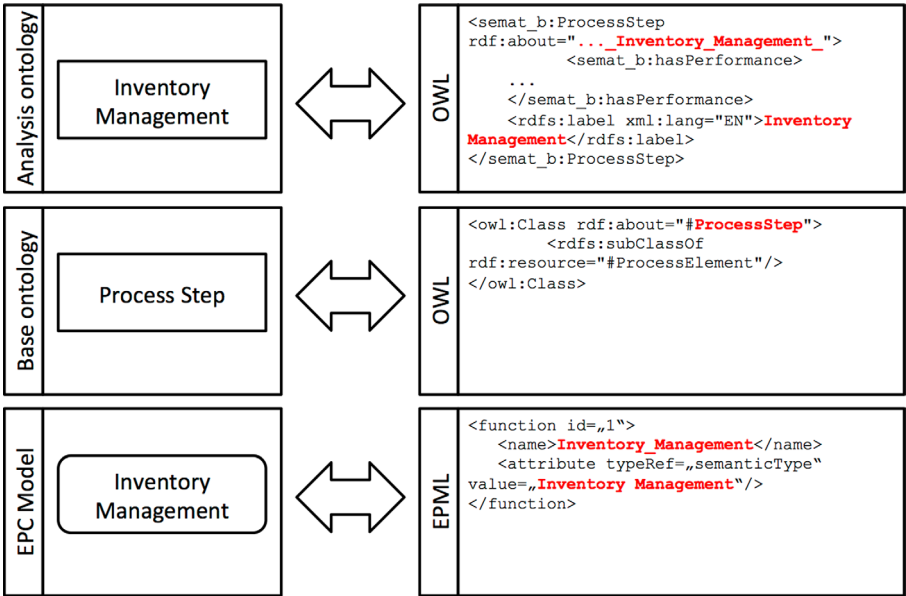


Figure 4. Representation of models and ontologies in EPML and OWL

In summary, the base ontology and domain ontology are used for semantic description. Furthermore, together with these, instances of the transferred model constructs form the analysis ontology (Figure 4). All analyses and comparisons of models are only based on this ontology. This ontology thus comprises all the information that will be used in further analyses (Table II) in benchmarking initiatives. The analysis ontology contains information ($C_{n,p}$) about each model construct n (the process itself is also considered a model construct) for every process model p . Additionally, it contains m performance information ($PI_{n,p,m}$), or other relevant process aspects, and context information of all model constructs and of the underlying k domain concepts' mapping type ($M_{n,p,k}$).

Step 6: analyzing the analysis ontology. The benchmarking expert analyzes the instance ontology by using ontology query languages, such as the SPARQL Protocol And RDF Query Language (SPARQL) (Sirin and Parsia, 2007), and by using predefined benchmarking metrics.

4.2 Creating an analysis ontology: an illustrative example

The following example illustrates our benchmarking approach's process steps by semantically comparing two process models from the logistics domain. The Process Classification FrameworkSM (APQC, 2008) taxonomy is used as the domain ontology. This taxonomy allows objective performance comparisons of cross-functional business processes within and between organizations. Furthermore, we use the base ontology presented in Section 4.1.

To illustrate the mapping of process elements to the domain and base ontology, we use two different process models that represent a process of this domain. One of the processes is modeled in UML and the other in EPC. The latter applies to only three element types: function, event, and control flow. The first is an activity diagram and uses action, control flow, and object nodes as element types. The process model is transferred to the analysis ontology through an automated instantiation of the model elements as instances of the base ontology concepts. This can be inferred from the instantiation of the "Inventory Management" function within the EPC in our illustrated example (arrow pointing sideways from "Inventory Management" as an element of the EPC towards "Inventory Management" as an element of the analysis ontology) in Figure 5.

The example in Figure 5 reveals that two element types, namely "function" (EPC) and "action" (UML) are syntactically equivalent and are transferred as instances of the concept "Process Step." We can draw further conclusions when the generated instances of the analysis ontology are mapped to the domain ontology's instances (illustrated by arrows

Table II.
Information in the
analysis ontology

Model constructs	Performance information				Domain concepts			
	PI_1	PI_2	...	PI_m	DC_1	DC_2	...	DC_k
$C_{1,1}$	$PI_{1,1,1}$	$PI_{1,1,2}$	...	$PI_{1,1,m}$	$M_{1,1,1}$	$M_{1,1,2}$	...	$M_{1,1,k}$
$C_{2,1}$	$PI_{2,1,1}$	$PI_{2,1,2}$	...	$PI_{2,1,m}$	$M_{2,1,1}$	$M_{2,1,2}$	...	$M_{2,1,k}$
...	...	...	...	...	...	...	...	...
$C_{n,1}$	$PI_{n,1,1}$	$PI_{n,1,2}$	...	$PI_{n,1,m}$	$M_{n,1,1}$	$M_{n,1,2}$	...	$M_{n,1,k}$
$C_{1,2}$	$PI_{1,2,1}$	$PI_{1,2,2}$	...	$PI_{1,2,m}$	$M_{1,2,1}$	$M_{1,2,2}$	...	$M_{1,2,k}$
$C_{2,2}$	$PI_{2,2,1}$	$PI_{2,2,2}$	...	$PI_{2,2,m}$	$M_{2,2,1}$	$M_{2,2,2}$	...	$M_{2,2,k}$
...	...	...	...	...	...	...	...	...
$C_{l,2}$	$PI_{n,2,1}$	$PI_{n,2,2}$	...	$PI_{n,2,m}$	$M_{n,2,1}$	$M_{n,2,2}$	...	$M_{n,2,k}$

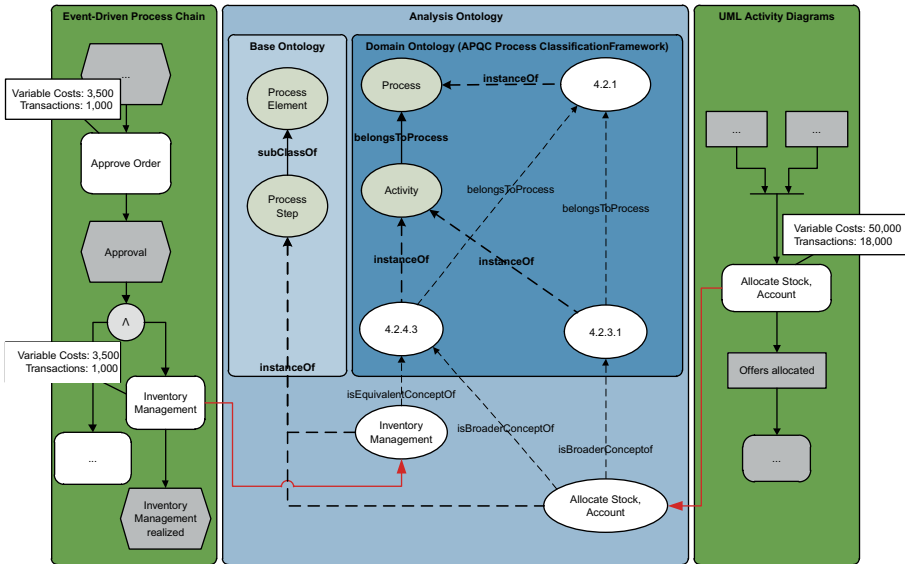


Figure 5.
Example

labeled “isEquivalentConceptOf” or “isBroaderConceptOf” in Figure 5). Even if the mapping has been done by different people, we can still conclude that a part of the “Allocate Stock, Account” process step of the UML activity diagram is identical to an EPC process step (“Inventory Management”). If additional performance information (e.g. costs and transactions) and context information on the individual process steps are included, we can draw further conclusions. We can also deduce that, as part of the UML activity diagram, information on the costs of the “Allocate Stock, Account” construct must be separated to make it comparable to the “Inventory Management” construct within the EPC.

5. Evaluation

The evaluation of IT artifacts (i.e. methods and software prototypes) on the basis of design-oriented research approaches is problematic from a methodological point of view (Frank, 2006, pp. 5-6; Vanderhaeghen *et al.*, 2010, p. 25). The outcome of design-oriented research should be evaluated in as many “real” contexts (real-world scenarios) as possible, although this is hardly fully possible in the research project context (Vanderhaeghen *et al.*, 2010, p. 25). Owing to these problems, we follow a twofold evaluation approach, as outlined in Section 3: first, based on our prototype, we simulate process benchmarking in experiments. Second, we follow the proposal by Frank (2006, pp. 33-35) to evaluate design-oriented research argumentatively and descriptively. We do so specifically for the economic evaluation perspective. In the following sections, we will outline the results of our different evaluations.

5.1 Technical evaluation

Our primary objective with the technical evaluation is to prove the concept’s general implementability by means of a software system. In a detailed description, we will elaborate on the means by which the semantic analysis of process models can be implemented efficiently. In order to achieve our objective, we developed an experimental

prototype called SEMAT Analysis Tool (SEMantic ATtribution). This tool uses a cyclic process to develop and test design alternatives. The most current version of this prototype makes use of the SEMAT Framework, which contains the proposed approach’s main functionalities (Figure 6). This framework is also used by the SEMAT Ontology Editor Plugin, which is utilized to create domain ontologies, as well as the SEMAT Modeling Tool Plugin to model and attribute the process models. A repository in the form of a database system was created to store all results centrally, as well as to recover and further process them. Since the technical challenge was to provide an automated analysis of process models, an in-depth description of all the components would thus be beyond the scope of this article; consequently, our discussion focuses on the SEMAT Analysis Tool.

The SEMAT Analysis Tool has been developed by means of Java, which is used for the analysis and comparison of process models. Models can be loaded from the repository or imported from a comma separated (CSV) or EPML file (Mendling and Nüttgens, 2006). In order to conduct the analysis of the process models within a single ontology, imported models are converted into an OWL 1.1 (W3C, 2006) analysis ontology (see Section 4.1). In order to assure the evaluation is made on the basis of a valid analysis ontology, analysis ontologies can be validated in advance with regard to soundness, completeness, decidability, and correctness. The prototype provides two alternative methods of calculating metrics for the interpretation of analysis ontologies. Analysts can choose one or more precompiled metrics that suit their needs, or they can run custom queries via SPARQL (Sirin and Parsia, 2007). Figure 7 contains a screenshot of the prototype – showing results from the example as discussed in Section 5.2.

To ensure a high degree of quality, an in-depth review was undertaken of the prototype’s implementation and test results. For instance, the analysis ontology creation was formally validated. The SEMAT Analysis Tool creates sound and determinable analysis ontologies if the domain ontologies are sound and determinable, and do not

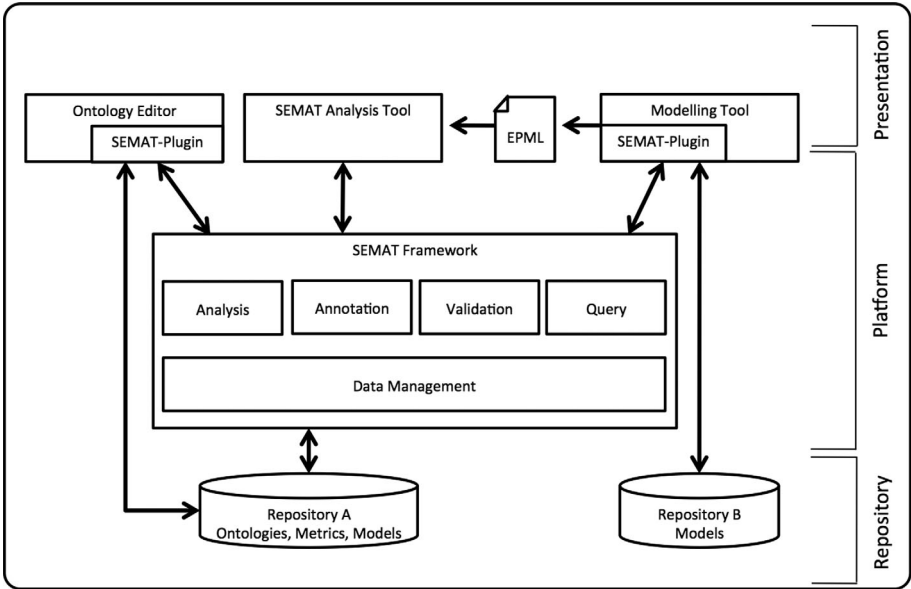
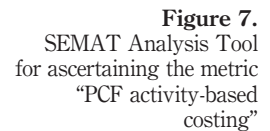


Figure 6.
SEMAT software
components



In conclusion, no technical limitations that could have influenced the general concept were found.

For the conceptual evaluation, we focused on procurement processes, using two prominent reference models, namely Scheer's reference model for industrial business processes (Scheer, 1994), and the Aachener PPS Model (Schuh, 2006). The decision to select these models was driven by the domain ontology selection. The conceptual evaluation's main objective is to show the practicability of our approach. In order to also support the technical evaluation results, we used the SEMAT prototype for this conceptual evaluation.

In order to determine the ideal problem domain, the Process Classification FrameworkSM (PCF) (APQC, 2008) taxonomy was transferred to a determinable OWL-DL-Ontology, and used as the domain ontology for further analysis (APQC – American Productivity Center and the American Productivity & Quality Center). The APQC and member companies developed the PCF as an open standard to facilitate improvement through process management and benchmarking, regardless of industry, size, or geography. The PCF organizes operating and management processes into 12 enterprise-level categories, including process groups, and over 1,500 processes and associated activities. The PCF, associated measures, and benchmarking surveys are available for download and completion at no charge at www.apqc.org/OSBCdatabase. The transformation of the entire PCF leads to four concepts (“Category”, “Process Group”, “Process”, and “Activity”) and 1,031 individuals (mostly activities).

Furthermore, for every PCF concept, an individual “No Match” was created. Unspecified reference model constructs could therefore also be mapped to the domain ontology.

We had to map all the process model elements to the domain ontology. We enriched Scheer’s (1994) reference model and the Aachener PPS Model (Schuh, 2006) with random performance information, including the number of transactions, variable costs, and fixed costs, as they might occur in real-world scenarios. This additional context information allows an activity-based costing and further analysis on the basis of several metrics. Table III illustrates the mappings to the PCF, as well as the assumed performance values. The prototype provides the use of the three mapping types mentioned in Section 4.1.

The first mapping procedure result showed that no exact mappings to the domain ontology could be allocated for all the models’ constructs. Some constructs were not enclosed in the domain ontology, which means that either the domain ontology has to be enhanced by including these constructs, or they must be excluded. In order to enhance the quality of the mappings, two authors mapped them independently. The inter-rater reliability (inter-rater percentage agreement) was approx. 95 percent. Thus, 5 percent of the mappings were subjected to a discussion to reach an agreement.

The SEMAT Analysis Tool provides some precompiled quantitative and qualitative metrics. For example, one activity-based costing metric is applied to process model’s activities, whereas others are applied to the domain ontology’s concepts.

In activity-based costing (Rosenkranz, 2006), the activity cost rate is calculated as the total cost per transaction. This is done by adding the marginal costs to the marginal costs multiplied by the total fixed costs divided by the total variable costs. The marginal costs are calculated by dividing the individual variable costs by the number of transactions. The metric, which is intended for the domain concepts, can be used for the comparison of models, because it ascertains activity-based costs for each PCF concept (the variables are described in Table II):

$$ABC(DC_{j,\phi}) = \frac{\sum_{i=1}^n (PI_{i,\phi,2} \cdot f(M_{i,\phi,j}))}{\sum_{i=1}^n PI_{i,\phi,3}} \cdot \left(\frac{\sum_{i=1}^n PI_{i,\phi,1}}{\sum_{i=1}^n PI_{i,\phi,2}} + 1 \right)$$

The ABC metric can be used for the quantitative analysis and the comparison of models. Like all other metrics here, its use relies on the assumption that one activity’s modeled performance information is evenly distributed to all the existing domain ontology subconcepts.

As an example, we created additional qualitative metrics by adapting metrics from a representational analysis of Recker *et al.* (2009). This analysis is based on the representation theory developed by Wand and Weber (Weber, 1997), which is in turn an adaptation of an ontology proposed by Bunge (1977). These adapted metrics are normally used to determine modeling techniques’ effectiveness and efficiency (Recker *et al.*, 2009), which can be achieved by measuring the models’ degree of conformity with the domain ontology. Thus, in combination with the ABC metric above, qualitative metrics can provide detailed insights into the models. The following four additional metrics were used for the example analysis:

$$(1) \text{ Degree of Completeness (DoC)} = \frac{\text{Number of modeled domain concepts}}{\text{Total number of domain concepts}}$$

	Fixed costs (PI ₁)	Variable costs (PI ₂)	Transactions (PI ₃)	APQC process classification framework													
				0.0.0.0 (DC ₁)	4.2 (DC ₂)	4.2.1.3 (DC ₃)	4.2.1.4 (DC ₄)	4.2.1.5 (DC ₅)	4.2.3.1 (DC ₆)	4.2.3.3 (DC ₇)	4.2.3.4 (DC ₈)	4.2.3.5 (DC ₉)	4.2.3.6 (DC ₁₀)	4.2.3.7 (DC ₁₁)	4.2.4.2 (DC ₁₂)	4.2.4.3 (DC ₁₃)	4.2.4.4 (DC ₁₄)
Aachener PPS (C _{1,1})	4,500	–	–	1													
Order quantity calculation (C _{2,1})	–	6,500	1,000			2	2	2									
Preparation of requests (C _{3,1})	–	8,000	1,000						3								
Evaluate proposal (C _{4,1})	–	4,500	1,000						3								
Select supplier (C _{5,1})	–	6,000	1,000						3								
Approve order (C _{6,1})	–	3,500	1,000							1							
Monitor order (C _{7,1})	–	3,500	1,000								2	2	2	2	2	1	2
Inventory management (C _{8,1})	–	3,000	1,000		1												
Scheer (C _{1,2})	200,000	–	–														
Develop requisition, allocation cost unit, cost unit accounting (C _{2,2})	–	60,000	20,000	2					2		1						
Request supplier (C _{3,2})	–	80,000	10,000														
Update basic data (C _{4,2})	–	15,000	10,000	1													
Select supplier and order quantity, account supplier, impersonal account, create pro forma invoice (C _{5,2})	–	70,000	20,000	2			2			2							
Printing of purchase orders and transmission (C _{6,2})	–	60,000	20,000									1					
Monitor order (C _{7,2})	–	120,000	20,000										1				
Create dunning (C _{8,2})	–	40,000	3,000	1											2		2
Quality testing (C _{3,2})	–	180,000	12,000														
Create reclamation, correct data (C _{10,2})	–	40,000	2,000	2													
Account receipt (C _{11,2})	–	20,000	18,000	1													
Allocate stock, account (C _{12,2})	–	50,000	18,000	2													
Invoice verification (C _{13,2})	–	100,000	18,000	1												2	
Correct invoice (C _{14,2})	–	40,000	1,000	1													
Account invoice (C _{15,2})	–	60,000	18,000	1													

Table III.
Mapping and performance information of the reference models

- (2) Degree of Redundancy (DoR)
= $\frac{\text{Number of model constructs that refers to the same domain concept}}{\text{Total number of model constructs}}$
- (3) Degree of Overload (DoO)
= $\frac{\text{Number of model constructs that refers to more than one domain concept}}{\text{Total number of model constructs}}$
- (4) Degree of Excess (DoE)
= $\frac{\text{Number of model constructs that refers to no domain concept}}{\text{Total number of model constructs}}$

The evaluation results are illustrated in Table IV. The degree of excess accounts for all identified model constructs not mapped in the domain ontology. For example, the authors could not find a representation in the domain ontology used (APQC) for 42.86 percent of the constructs in Scheer’s model. However, Scheer’s model depicts only one third of all concepts included in the domain ontology (DoC). In contrast, the Aachener PPS covers a larger part of the domain, but shows a 42.86 percent Degree of Redundancy (DoR). In addition, under the assumed conditions, some domain constructs in Scheer’s model cause very high (18.21) or no costs (0). Therefore, a comparison with the Aachener PPS and a detailed analysis of the respective processes seem reasonable.

In summary, the described automated model analyses can significantly accelerate and facilitate the process benchmarking procedure. A detailed analysis and comparison of the models are possible and can be easily repeated. This allows the benchmarking partners to establish regular benchmarking and to review process adjustments.

Metrics	Scheer	Aachener PPS
<i>Adapted Metrics (Recker et al., 2009)</i>		
Degree of Completeness (DoC)	33.33%	40.74%
Degree of Redundancy (DoR)	14.29%	42.86%
Degree of Overload (DoO)	14.29%	28.57%
Degree of Excess (DoE)	42.86%	0.00%
<i>ABC</i>		
4.2.1.3 Develop inventory strategy (10283)	0.00	7.34
4.2.1.4 Match needs to supply capabilities (10284)	4.25	7.34
4.2.1.5 Analyze company’s spend profile (10285)	0.00	7.34
4.2.3.1 Process/review requisitions (10292)	3.64	6.96
4.2.3.4 Create/distribute purchase orders (10295)	6.98	3.95
4.2.3.5 Expedite orders and satisfy inquiries (10296)	3.64	0.00
4.2.3.6 Record receipt of goods (10297)	7.28	3.95
4.2.3.7 Research/resolve exceptions (10298)	18.21	3.95
4.2.4.2 Prepare/analyze procurement and vendor performance (10300)	0.00	3.95
4.2.4.3 Support inventory and production processes (10301)	3.37	3.39
4.2.4.4 Monitor quality of product delivered (10302)	18.21	3.95

Table IV.
Evaluation results

5.3 Economic evaluation

The benchmarking approach proposed in this paper will only pay off if the costs of process model annotation are lower than the benefits gained from improved benchmarking efficiency and effectiveness.

Two business cases are presented for a clarification and an in-depth analysis of these facts. These can be obtained from the authors of this paper; they include an advanced Excel tool for cost-benefit calculation, thus enabling an effortless transfer of the expected cash flows to real process benchmarking projects based on generally accepted cost and benefit aspects (Table V). In the first business case A1 (alternative 1), “manual” process benchmarking is carried out, i.e. without implementing the method presented in this paper. The presented semantic process benchmarking is applied in the second business case A2 (alternative 2). The expected costs and benefits of the two alternatives are split into one-off and recurring elements (Table V), so that changes in single elements can be evaluated by means of a sensitivity analysis with regard to their effect on the applied metrics such as the net present value (Table VI). In order to make a risk assessment, the two business cases A1 and A2 are additionally evaluated according to pessimistic, realistic, and optimistic scenarios.

The expected benefits for the respective financial years according to the three scenarios’ current benchmarking are defined as follows (Table V):

- (1) *Pessimistic (p)*. The first benchmarking realizes a process cost reduction (“Original benefits” in Table V) of 50,000 EUR, the second another 40,000 EUR and the third another 30,000 EUR.
- (2) *Realistic (r)*. The first benchmarking realizes a process cost reduction of 60,000 EUR, the second another 50,000 EUR and the third another 40,000 EUR.
- (3) *Optimistic (o)*. The first benchmarking realizes a process cost reduction of 70,000 EUR, the second another 60,000 EUR and the third another 50,000 EUR.

The observation of the benefits continues up to three years after the last process benchmarking (in the subsequent years $t = 4$ and $t = 5$ no further cost reductions are realized in our business cases A1 and A2), so that the project duration at the initial investment in financial year $t = 0$ amounts to a total of five years (to $t = 5$).

The internal rate of return is set at 15 percent. Compared to alternative 1, a 15 percent higher investment cost (in total) is assumed for the first benchmarking by deploying the presented semantic process benchmarking method (alternative 2, see Table V, one-off-costs 1.6 to 1.9), and a 7 percent higher investment cost (in total) for the subsequent years (see Table V, running costs 2.1 to 2.3). The applied cash flows (payments and deposits) for expected costs and expected benefits (average values) are based on figures that the paper’s authors have drawn from scientifically accompanied process benchmarking projects at companies of different sizes and from various industrial sectors.

Table V displays the cash flows for the years $t = 0$ to $t = 5$ with the codes p – pessimistic, r – realistic and o – optimistic.

Table VI displays the results of the two business cases A1 (no deployment of the method) and A2 (deployment of the method) for the three scenarios examined on the basis of the figures in Table V.

The results show that the use of the proposed method of semantic process benchmarking is very beneficial. A follow-up sensitivity analysis and analyses of other business cases have shown that this depends strongly on the one-off acquisition costs and

Table V.
Costs and benefits of
(semantic) process
benchmarking
(exemplary business
cases)

	Description	Unit	Alternative 1 ("manual" process benchmarking): cash flows from t = 0 to t = 5 in € (thousands)		Alternative 2 (semantic process benchmarking): cash flows from t = 0 to t = 5 in € (thousands)	
			3 scenarios		3 scenarios	
Benefits						
	1. One-off benefits					
	1.1 Cost reduction					
	Cost reduction if a benchmarking is repeated	€ (thousands)	p = [0,0,0,0,0,0]		p = [0,30,30,0,0,0]	
	Better comparison of results with previous results		r = [0,0,0,0,0,0]		r = [0,30,30,0,0,0]	
	Reduced time to integrate other benchmarking partners		o = [0,0,0,0,0,0]		o = [0,30,30,0,0,0]	
	Automatic analysis of improved processes by using available metrics					
	An intensive analysis is undertaken of the problem domain and explicitly documented. The semantic aspects of the processes are also explicitly taken into account and analyzed	€ (thousands)	p = [0,0,0,0,0,0]		p = [0,0,0,0,0,0]	
			r = [0,0,0,0,0,0]		r = [0,0,0,0,0,0]	
			o = [0,0,0,0,0,0]		o = [0,0,0,0,0,0]	
2. Recurring benefits						
2.1 Original benefits		€ (thousands)				
	Original benefits (e.g. cost reduction due to improved processes) presumed by all variants of process benchmarking		p = [0,50,90,120,120,120]		p = [0,50,90,120,120,120]	
			r = [0,60,110,150,150,150]		r = [0,60,110,150,150,150]	
			o = [0,70,130,180,180,180]		o = [0,70,130,180,180,180]	
2.2 Cost reduction		€ (thousands)				
	Recurring cost reduction due to the benchmarking-improved processes		p = [0,0,0,0,0,0]		p = [0,0,0,0,0,0]	
			r = [0,0,0,0,0,0]		r = [0,0,0,0,0,0]	
			o = [0,0,0,0,0,0]		o = [0,0,0,0,0,0]	
2.3 Cost avoidance		€ (thousands)				
	Recurring cost avoidance due to benchmarking-improved processes		p = [0,0,0,0,0,0]		p = [0,0,0,0,0,0]	
			r = [0,0,0,0,0,0]		r = [0,0,0,0,0,0]	
			o = [0,0,0,0,0,0]		o = [0,0,0,0,0,0]	
2.4 Identification of potential for additional improvements		€ (thousands)				
	Potential for improvements, which would otherwise have remained hidden, can be identified with the aid of predefined metrics		p = [0,0,0,0,0,0]		p = [0,5,9,12,12,12]	
			r = [0,0,0,0,0,0]		r = [0,6,11,15,15,15]	
			o = [0,0,0,0,0,0]		o = [0,7,13,18,18,18]	
2.5 Other		€ (thousands)				
	For example					
	Further external know-how is applied		p = [0,0,0,0,0,0]		p = [0,0,0,0,0,0]	
	Increased transparency of the analyses by means of an explicit description of the metrics		r = [0,0,0,0,0,0]		r = [0,0,0,0,0,0]	
	<i>Ad hoc</i> analysis by means of SPARQL queries to obtain new information		o = [0,0,0,0,0,0]		o = [0,0,0,0,0,0]	
	Obtaining new knowledge about processes through inference					

(continued)

Description	Unit	Alternative 1 "manual" process benchmarking: cash flows from t = 0 to t = 5 in € (thousands)		Alternative 2 (semantic process benchmarking): cash flows from t = 0 to t = 5 in € (thousands)	
<i>Costs</i>					
1. One-off costs					
1.1 Defining the benchmarking subject					
	Choice of processes to be researched	€ (thousands)	p = [5,5,5,0,0,0]	p = [5,5,5,0,0,0]	
			r = [5,5,5,0,0,0]	r = [5,5,5,0,0,0]	
			o = [5,5,5,0,0,0]	o = [5,5,5,0,0,0]	
1.2 Assessment of own processes	Assessing own processes and determining the performance data	€ (thousands)	p = [45,45,45,0,0,0]	p = [45,45,45,0,0,0]	
			r = [45,45,45,0,0,0]	r = [45,45,45,0,0,0]	
			o = [45,45,45,0,0,0]	o = [45,45,45,0,0,0]	
1.3 Choice "best in class"	Selecting appropriate benchmarking partners	€ (thousands)	p = [30,30,30,0,0,0]	p = [30,30,30,0,0,0]	
			r = [30,30,30,0,0,0]	r = [30,30,30,0,0,0]	
			o = [30,30,30,0,0,0]	o = [30,30,30,0,0,0]	
1.4 Comparison of performance data	Comparing the performance data with the performance data of all the benchmarking partners' processes	€ (thousands)	p = [15,15,15,0,0,0]	p = [15,15,15,0,0,0]	
			r = [15,15,15,0,0,0]	r = [15,15,15,0,0,0]	
			o = [15,15,15,0,0,0]	o = [15,15,15,0,0,0]	
1.5 Design of the conversion	Developing and planning measures to improve these processes by means of the identified potential improvements	€ (thousands)	p = [5,5,5,0,0,0]	p = [5,5,5,0,0,0]	
			r = [5,5,5,0,0,0]	r = [5,5,5,0,0,0]	
			o = [5,5,5,0,0,0]	o = [5,5,5,0,0,0]	
1.6 Investments	Licensing and hardware costs	€ (thousands)	p = [0,0,0,0,0,0]	p = [2,0,0,0,0,0]	
			r = [0,0,0,0,0,0]	r = [2,0,0,0,0,0]	
			o = [0,0,0,0,0,0]	o = [2,0,0,0,0,0]	
1.7 Labor expenditure and cost of services	Training costs, consultancy, software and hardware customization, installation	€ (thousands)	p = [0,0,0,0,0,0]	p = [3,0,0,0,0,0]	
			r = [0,0,0,0,0,0]	r = [3,0,0,0,0,0]	
			o = [0,0,0,0,0,0]	o = [3,0,0,0,0,0]	
1.8 Initial annotation of process models	Costs arising from the initial annotation of the process models	€ (thousands)	p = [0,0,0,0,0,0]	p = [5,0,0,0,0,0]	
			r = [0,0,0,0,0,0]	r = [5,0,0,0,0,0]	
			o = [0,0,0,0,0,0]	o = [5,0,0,0,0,0]	

(continued)

Table V.

Table V.

	Description	Unit	Alternative 1 ("manual" process benchmarking): cash flows from t = 0 to t = 5 in € (thousands)	Alternative 2 (semantic process benchmarking): cash flows from t = 0 to t = 5 in € (thousands)
1.9 Acquisition or construction of the domain ontology	Expenditure on an agreed domain ontology and sharing this with all the benchmarking partners	€ (thousands)	p = [0,0,0,0,0,0] r = [0,0,0,0,0,0] o = [0,0,0,0,0,0]	p = [5,0,0,0,0,0] r = [5,0,0,0,0,0] o = [5,0,0,0,0,0]
2. Running costs				
2.1 Hardware and application costs	Operating the hardware, updates (repository, analysis tool, etc.)	€ (thousands)	p = [0,0,0,0,0,0] r = [0,0,0,0,0,0] o = [0,0,0,0,0,0]	p = [0,1,1,0,0,0] r = [0,1,1,0,0,0] o = [0,1,1,0,0,0]
2.2 External consultancy costs	As it cannot be assumed that ontology know-how is available, it usually needs to be purchased	€ (thousands)	p = [0,0,0,0,0,0] r = [0,0,0,0,0,0] o = [0,0,0,0,0,0]	p = [0,3,3,0,0,0] r = [0,3,3,0,0,0] o = [0,3,3,0,0,0]
2.3 Maintaining annotated process models and metrics	Update of the previous process customizations. Maintaining all the benchmarking partners' process models	€ (thousands)	p = [0,0,0,0,0,0] r = [0,0,0,0,0,0] o = [0,0,0,0,0,0]	p = [0,3,3,0,0,0] r = [0,3,3,0,0,0] o = [0,3,3,0,0,0]

the process models' semantic annotation, as well as on the obtained benefits. In business cases in which no repeated process benchmarking was undertaken, the use of the procedure did not always prove advantageous, since the one-off costs exceeded the benefits.

Although the annotation of process models can be partly automated by using algorithms that match semantically (Giunchiglia *et al.*, 2007; Guarino, 1997), it is still a time-consuming task that experts need to carry out. It is therefore reasonable to search for application scenarios in which the relatively high costs of model preparation and transformation are outweighed by a large number of applications of the respective models.

We believe that a clearing centre that regularly carries out benchmarking projects for one or more specific domains, will best implement our approach. Such a clearing centre will be responsible for collecting, annotating, transforming, and storing the process models. It will offer interested parties benchmarking services by comparing their process representations with those that have already been processed and transformed. Consequently, the relatively high costs of model preparation are divided between a large number of organizations that utilize the benchmarking data. Moreover, developing and maintaining an adequate ontology and the associated software systems will then be affordable and can be organized centrally. The specific value proposition of the business model described here is therefore that the benchmarking offering is based on a large number of benchmarking participants and an automated, but rich, qualitative analysis that is also relatively cheap, because it does not require experts. We assume that the income generated by benchmarking offerings will directly depend on the number of benchmarks available, since benchmarking becomes more attractive if information from a larger number of organizations is available. Moreover, a large benchmarking database will simplify marketing activities and thus promote the acceptance of the services offered. In fact, each additional benchmarking participant makes the offering more attractive for all the other participants.

6. Conclusion

6.1 Contributions

In this article, we have proposed an approach whereby the time-consuming and costly analysis of process models for benchmarking will be partially automated, so that the performance indicators, as well as qualitative differences between process models, will become apparent. This approach contributes to practice and research in several ways. The theoretical contributions are threefold. First, we have proposed a novel and innovative approach to process model benchmarking, which makes this task more efficient and effective, and, simultaneously, more meaningful. Second, the developed SEMAT prototype, that demonstrates and proves the proposed approach's general viability, supports it efficiently. Third, we have defined fundamental, conceptual building blocks that can be used for further research. Furthermore, applying this approach allows novel and interesting information on adequate performance metrics to

Metric	Unit	Pessimistic		Realistic		Optimistic	
		A 1	A 2	A 1	A 2	A 1	A 2
Net present value	T EUR	56	110	132	194	207	277
Internal rate of return	%	26	36	39	50	52	63
Payback	Years	3.33	2.87	2.87	2.50	2.56	2.25

Table VI.
Results (exemplary
business cases)

be generated. This can be done by applying benchmarking analyses of process models that refer to different prominent reference models (e.g. Aachener PPS and Scheer's reference model for industrial business processes). Table VII summarizes the main contributions of our research to research and practice.

6.2 Limitations

Since all research has its limitations, we have also identified our approach's limitations, which thus point out directions for further research (see next Section 6.3). The proposed approach depends strongly on the availability of appropriate metrics and ontologies, as well as on the annotation of these ontologies to process models, which is a time-consuming task. The field of business administration still lacks organizations that set standards to develop and standardize ontologies for benchmarking purposes. Most available tools for ontology engineering and annotation are prototypes. Thus, they are not mature enough to enable easy and cost-effective ontology engineering and annotation in the context of semantic process benchmarking initiatives.

Evaluation perspective	Main questions	Evaluation methodology	Main contributions
Technical	Is the concept technologically feasible, and are there any technical limitations? Is it possible to implement the concept efficiently?	Development of a prototype and an in-depth review of the implementation	Software systems can support the concept efficiently The prototype creates sound and determinable analysis ontologies No technical limitations that could have had an influence on the general concept were found
Conceptual	Is the concept useful? Does it produce novel and interesting information?	Simulation based on metrics applied to several prominent reference models	The approach can generate novel and interesting information about processes, for example, the process costs of a domain concept that refers to different models The mapping procedure is the most time-consuming part of the concept. In-depth knowledge is required of the activities and the interpretation of their domain concepts and model constructs' meanings Advanced ontology know-how and programming skills are not required to generate new information in order to use precompiled metrics The development of a semantic process query language (SPQL) to facilitate the benchmarking analysis is crucial, and necessitates additional research
Economic	Can the concept be implemented profitably?	Conceptual analysis	The approach is too expensive to be implemented with only a few benchmarking participants If large benchmarking clearing centers are established, the approach will be more cost-effective

Table VII.
Main contributions

In addition, control-flow-related problems have to be solved. For example, since process models may involve cycles, decision points, and complex routing, it is difficult to determine the order in which parts of the process should be carried out.

6.3 Future research

The proposed approach can change the way process model benchmarking is carried out. However, in-depth knowledge of ontology engineering and information modeling is necessary to implement the approach, and an adequate tool support is required. The approach is viable, but for real-world scenarios, an advanced synthesis of the results obtained from the automatic analysis would further increase the relevance for practice, for example, by automatically creating meaningful benchmarking results. Furthermore, in the economic analysis outlined in Section 5.3, we have only taken a first step in researching this field. Empirical findings from the proposed approach's application in real-world scenarios would support the validation of the propositions underlying this research. To further strengthen the evaluations of the approach and the prototype, we also plan to conduct additional experiments, lab studies, and action research to determine their applicability and usability even better.

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